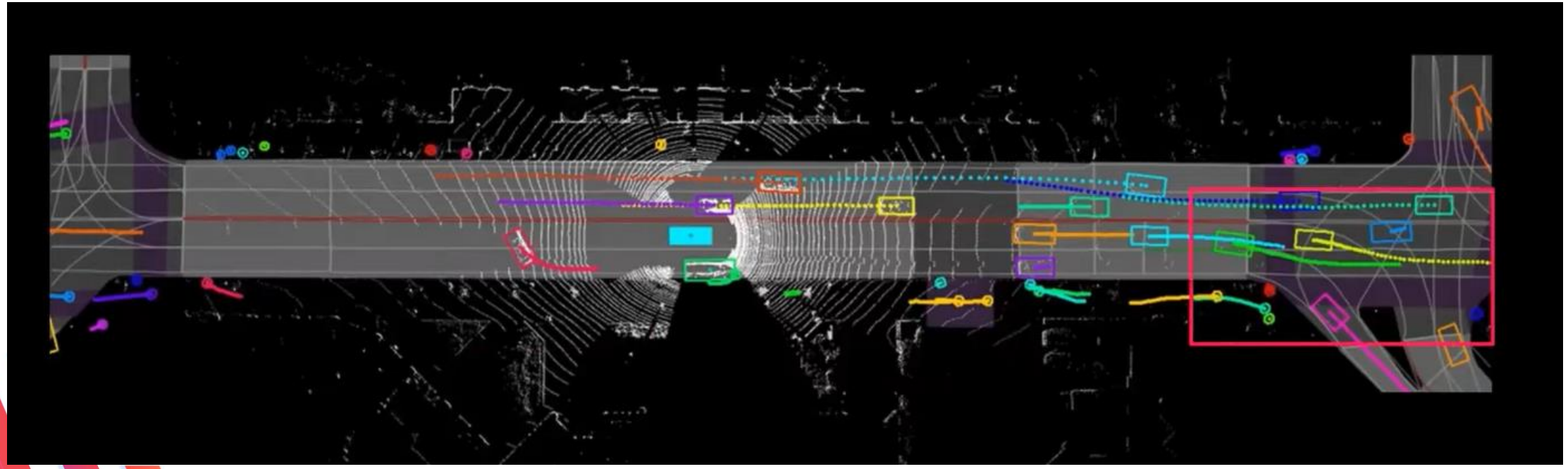
香港城市大學
City University of Hong Kong
Innovating into the Future

Chapter 4: Prediction

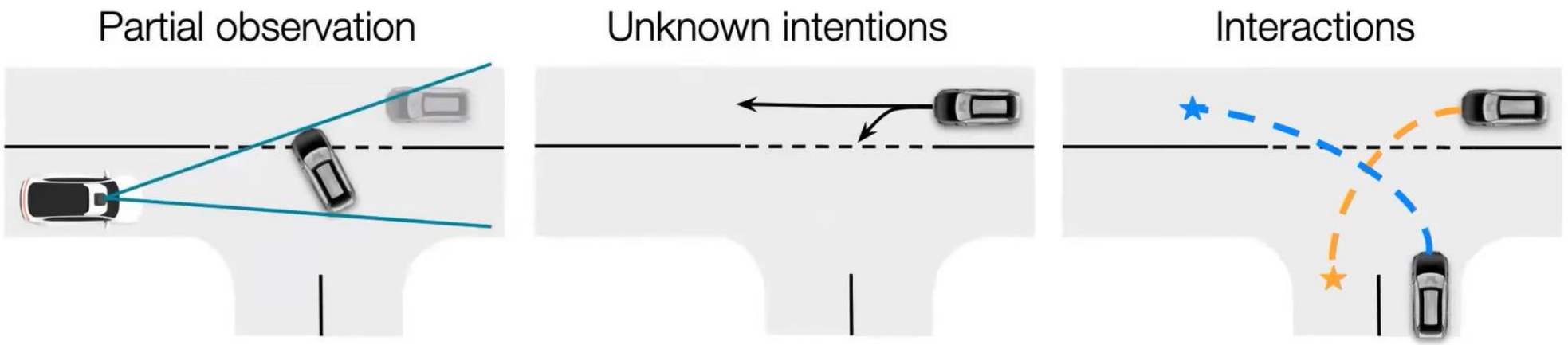
Prediction

Maps
Sensors
→
Perception
Detections Tracks
→
Prediction
Long Term Predictions
→
Planning
Motion Trajectory
→
Control
Steering / Acceleration



Main Challenges

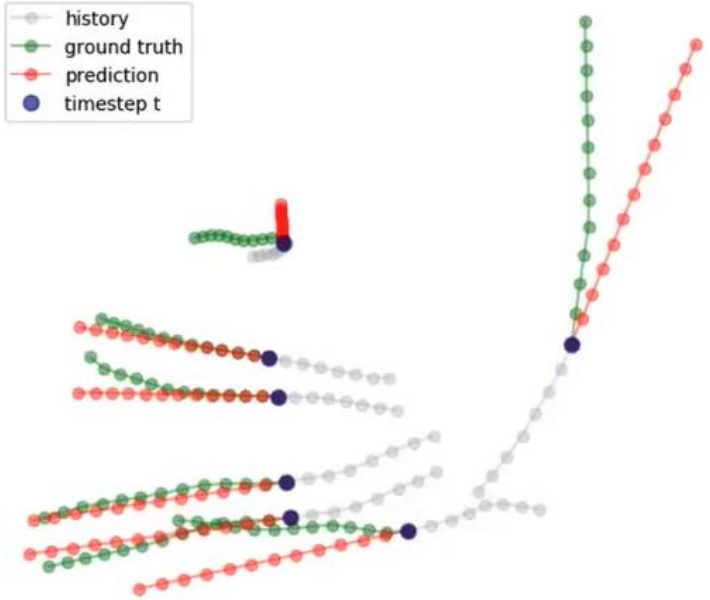
- Partial Observation: The vehicle can only perceive a subset of the environment, due to sensor limits (e.g., occlusions, field-of-view constraints)
- Unknown Intentions: Human drivers have *unobservable goals* (e.g., lane change, turn, stop) that the vehicle must infer from sparse observations
- Interactions: Road agents (cars, pedestrians, cyclists) do not act in isolation—their behaviors are *mutually dependent*



Kinematic Approach

- Predict future state using kinematic models, e.g.,
  - Constant Velocity (CV)
  - Constant Turn Rate and Acceleration (CTRA)

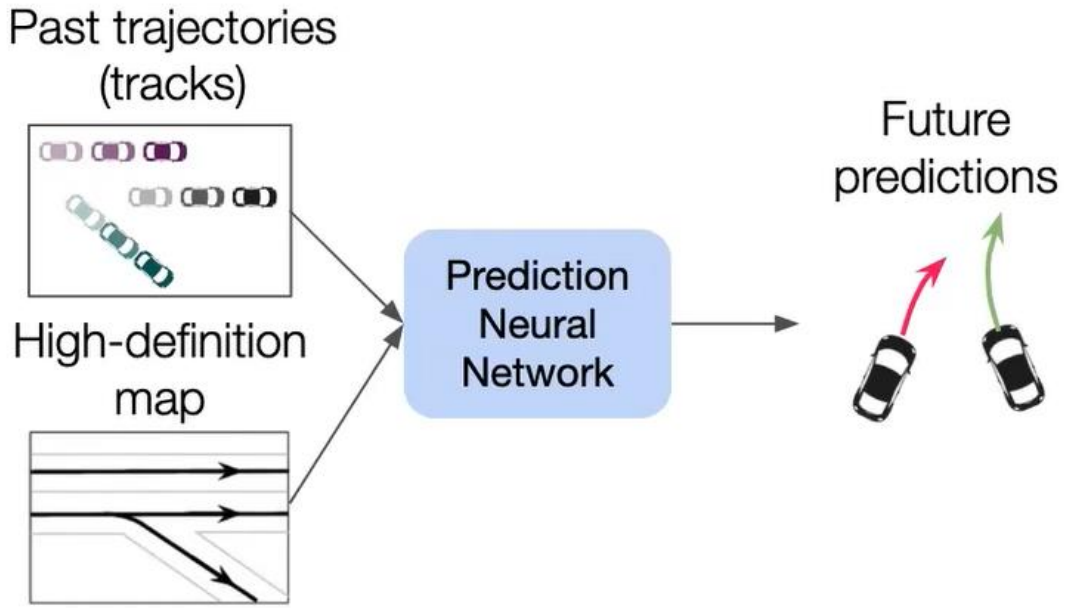
- Strengths
  - Incorporate prior knowledge about typical behavior and physics
  - Simple and fast
- Weaknesses
  - Cannot capture diversity of real world
  - Safety-critical cases do not follow standard behavior



Data-driven Track-based Prediction

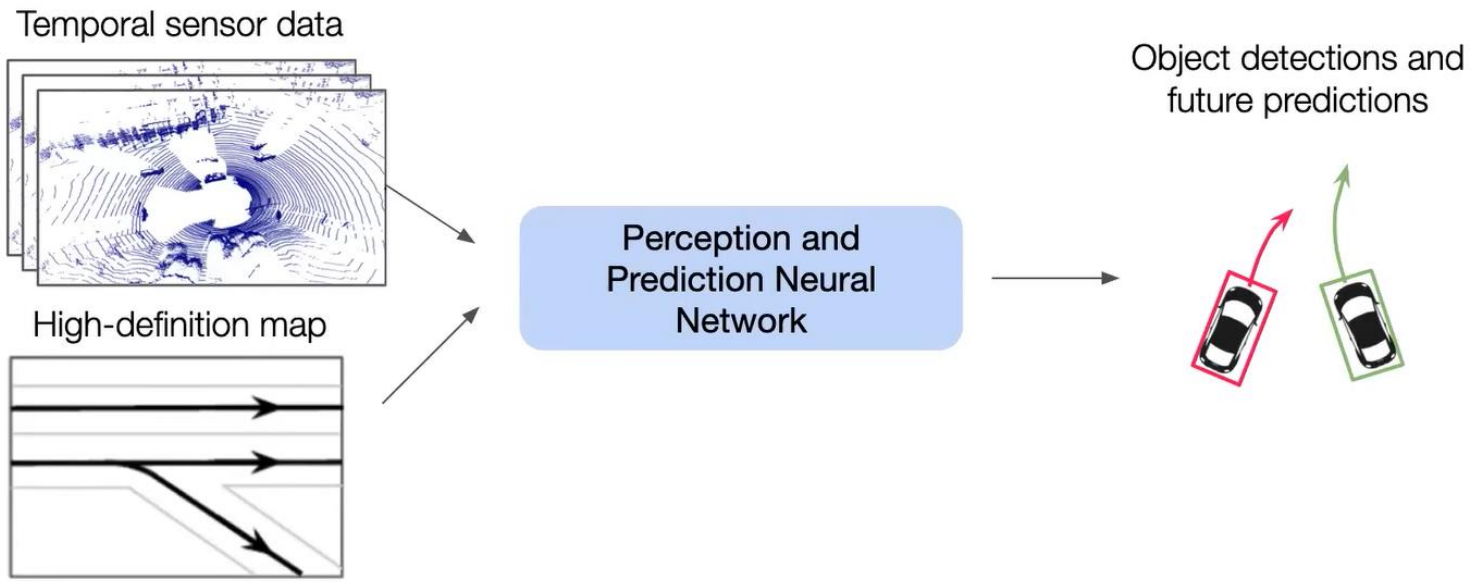
- Given past trajectories (obtained from Tracking) and an HD map
- Neural networks learn predictive patterns directly from data, rather than relying on hand-engineered knowledge

- Limitations:
  - Noisy detection and tracks
  - High computational cost
  - Information loss
  - Low explainability



Joint Perception and Prediction (PnP)

- Motion forecasting directly from sensor data (LiDAR) and HD Maps
- Advantages:
  - Interpretable
  - **Propagates uncertainty downstream**
  - Shared computation



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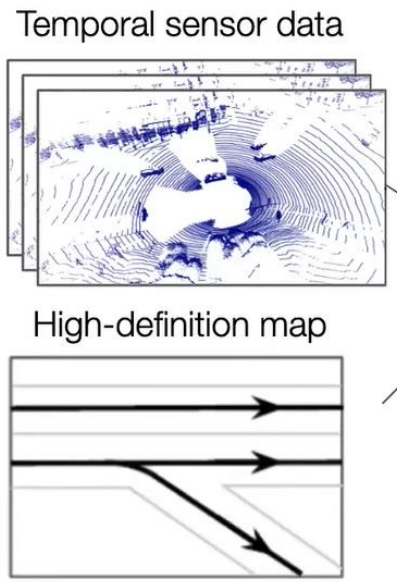


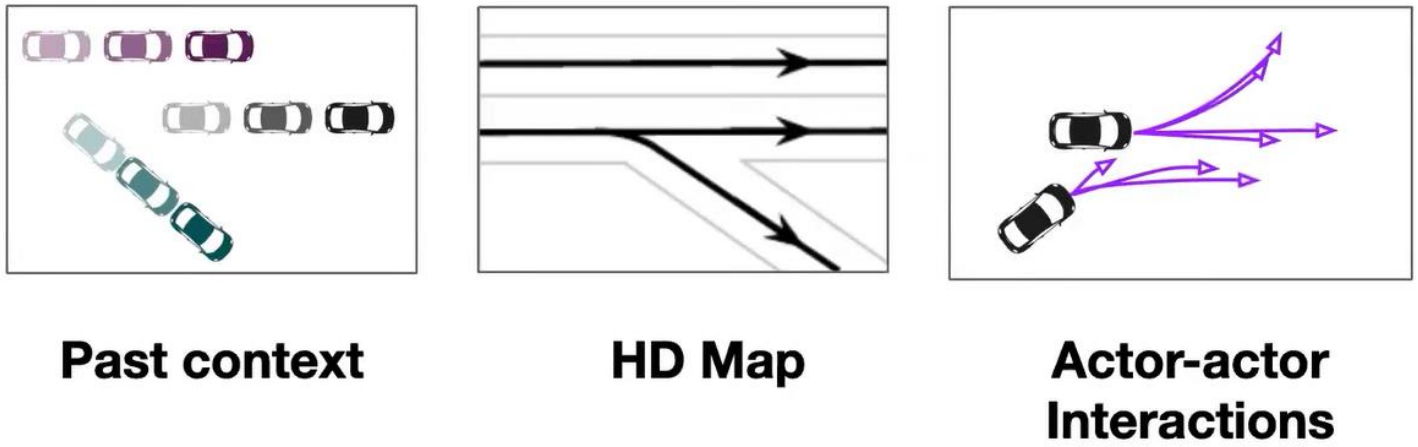
Why using tracking cannot well propagate uncertainty?

- The curse of dimensionality: too much information to maintain
- Information bottleneck: cannot keep the source of uncertainty
- Temporal correlation is lost: uncertainty is highly correlated over time

Joint Perception and Prediction (PnP)

- Input of PnP: raw sensor data + scene representation

- Scene representations:



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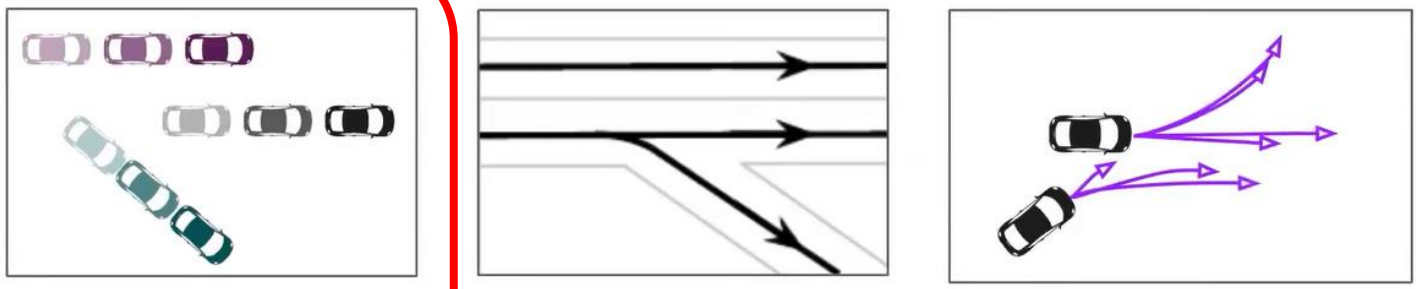


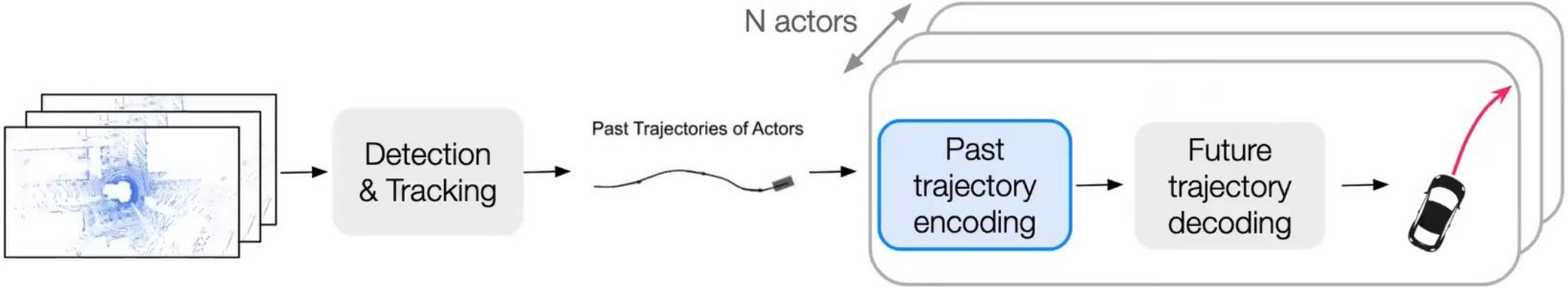
Past context

HD Map

Actor-actor Interactions

Extracting Past Context from Tracks

- Learned past trajectory encoder
  - A neural network that compresses the entire history of an agent's movement into a dense vector (embedding) that captures nuanced motion patterns
  - E.g., **RNN** (good for sequential data, model time steps one by one), or **transformers** (better at capturing long-range dependencies, e.g., how the car behaved 5 seconds ago affects its current intent)
  - Typically trained together with the decoder in an end-to-end manner



Encoding Tracks as Rasters

- Raster: a grid-based 2D image in BEV
  - Use 2D CNN to extract features
  - Multi-channel: separate and organize different types of semantic information, so the neural network can learn to interpret them effectively
  - Local to each actor: local coordinate system centered on the target vehicle, so the model learns motion patterns instead of location-specific ones

- Strengths
  - Simple to implement

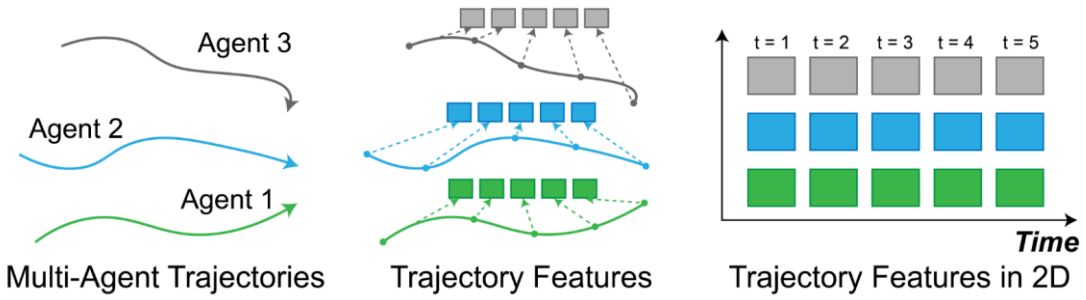
- Weaknesses
  - Memory intensive
  - Detection & tracking can be noisy

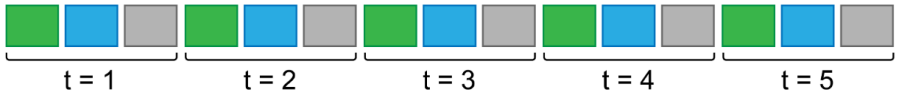
Encoding Tracks as Trajectories

- Sequence of waypoints or vectors
  - Single agent: extract features by, e.g., RNN/1-D CNN
  - Multiple agents: extract features by, e.g., Transformer

- Strengths
  - Compact representation

- Weaknesses
  - Past Trajectory has limited expressivity
  - Detection& tracking can be noisy





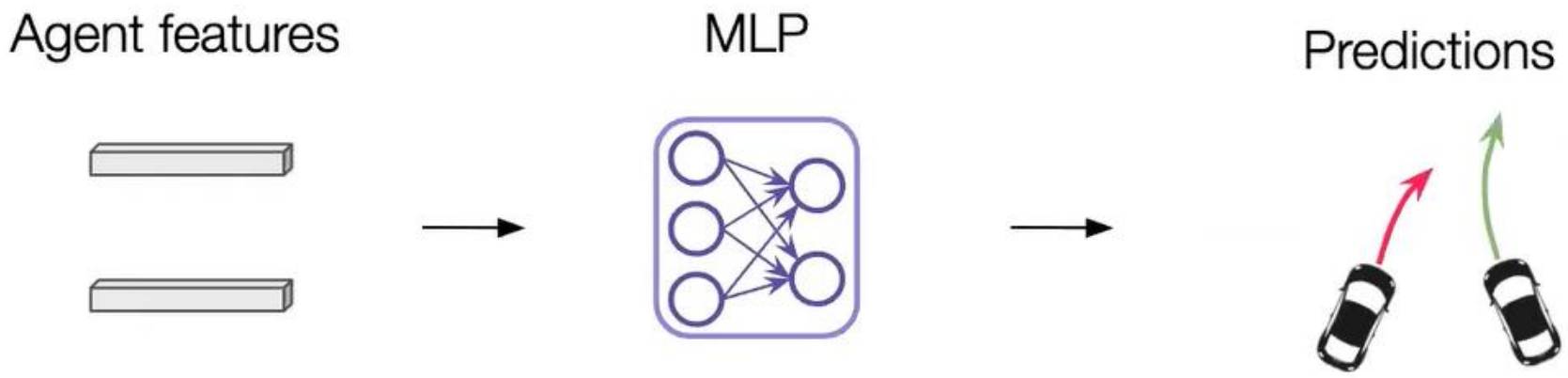
Agent-Aware Transformer
(Preserver Time & Agent Information)

Extracting Past Context from LiDAR

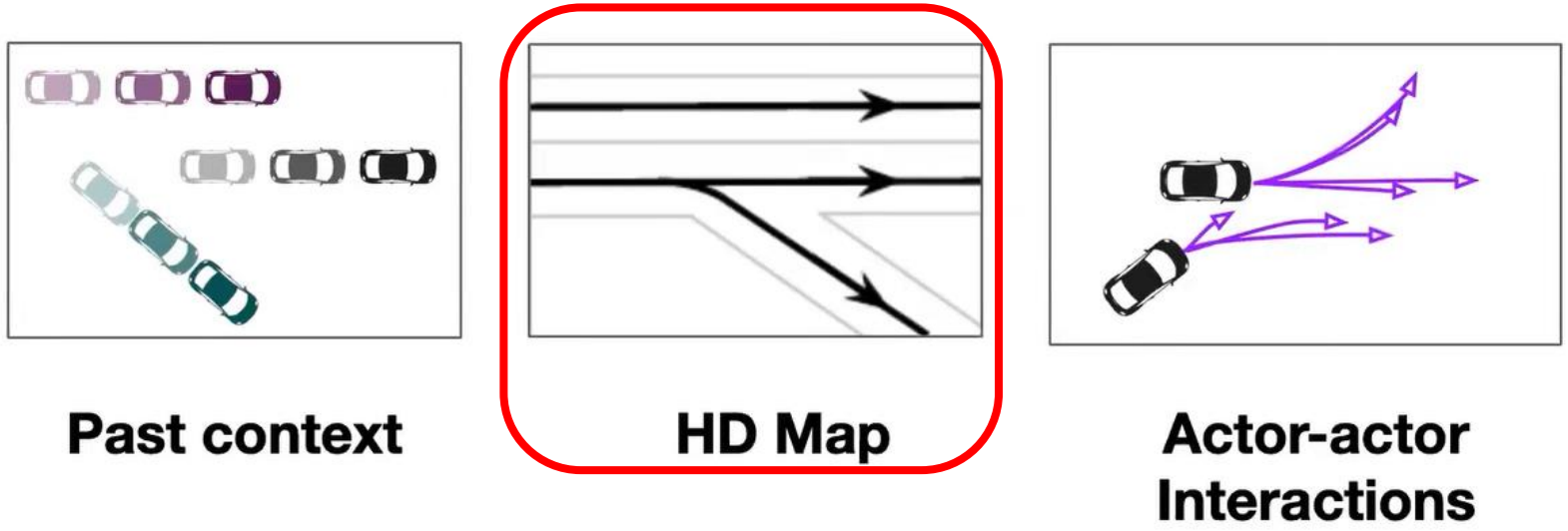
- Multiple LiDAR sweeps as input
  - Voxelize in current coordinate frame
  - Projected to BEV map to capture spatio-temporal information
- Extract agent features at detection locations
  - First run a lightweight detector to find candidate agent locations
  - For each detection, crop a small feature patch from the BEV map
  - A small CNN then processes this patch to extract a compact agent feature vector

Prediction

- From agent features, we can decode a trajectory as a vector
  - 2 coordinates: x, y waypoints
  - Several timesteps
- Decode the trajectories using a, .e.g., MLP, transformer or diffusion model
  - MLP: Multi-Layer Perceptron (a basic neural network with a few fully connected layers)

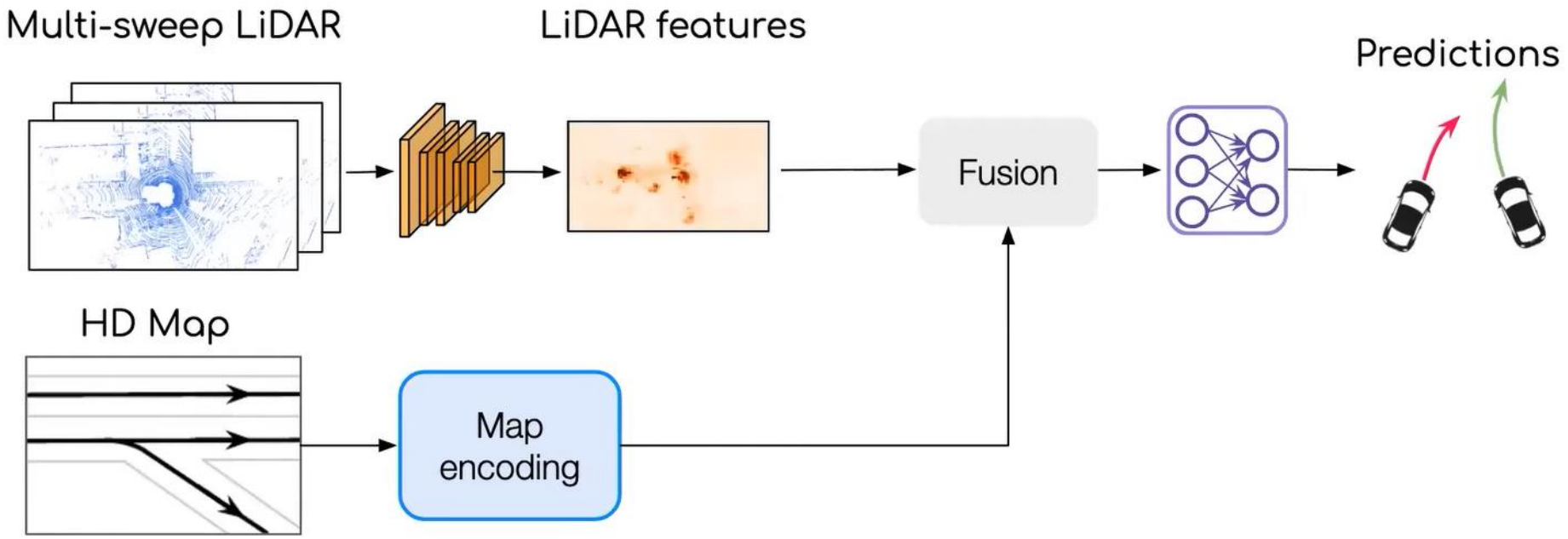


Representation of the Scene



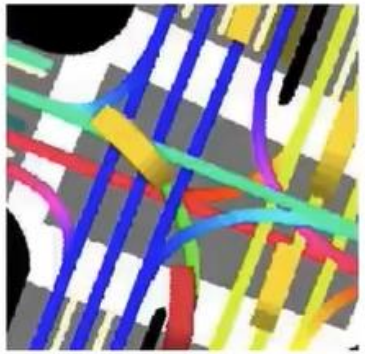
Encoding Map Context

- Add **map context** (lane lines, turn restrictions, stop signs)
  - Physically plausibility
  - Long-term prediction
  - Robustness

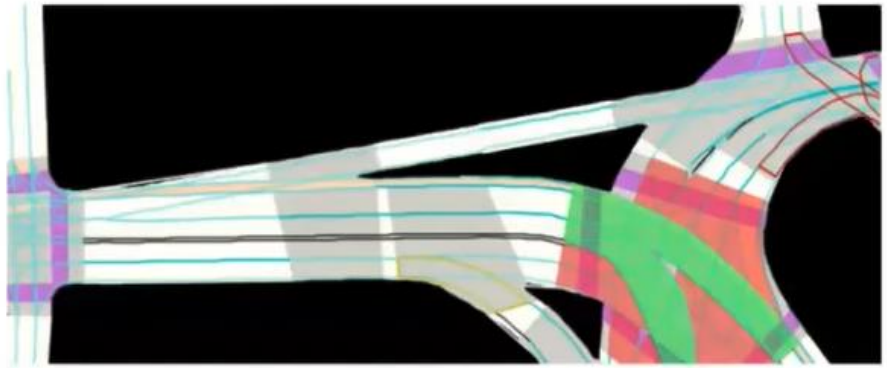


HD Map as Raster

- Raster representation
  - Use colors or masks to represent semantic elements

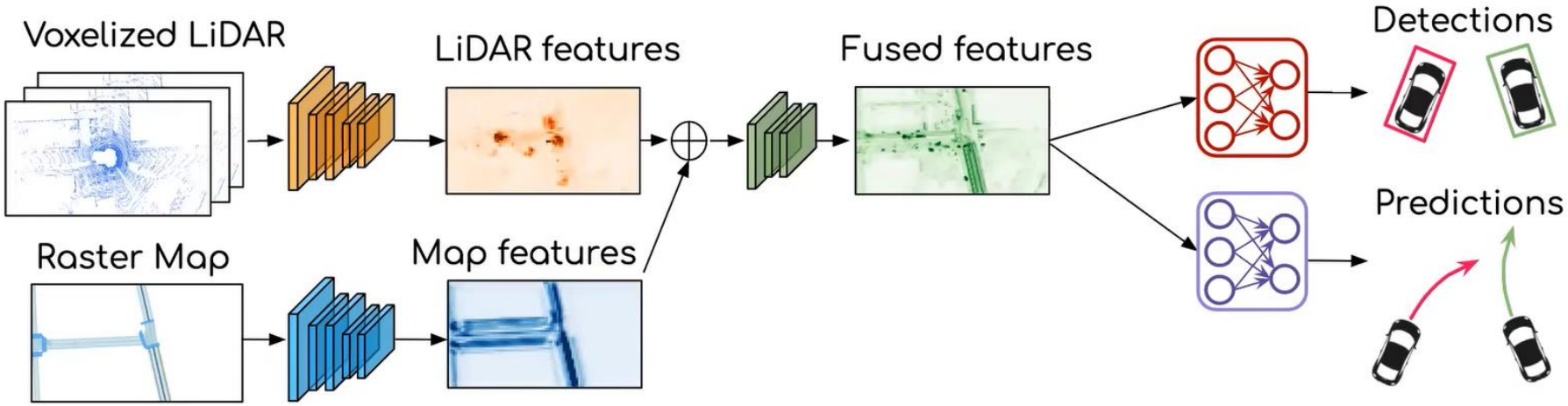


- Strengths
  - Simple to implement
  - Natural for dense prediction (e.g., occupancy)
- Weaknesses
  - Limited receptive field
  - Limited resolution
  - Memory intensive



LiDAR Context and Map Raster Fusion

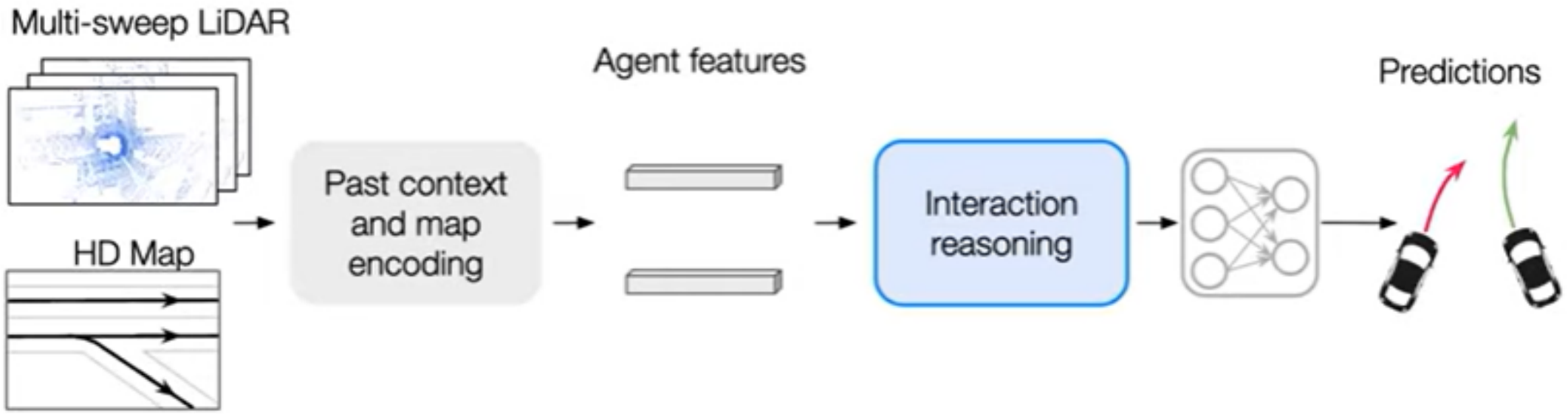
- Extract LiDAR and map features in a two-stream architecture
- Simple fusion by channel-wise concatenation of feature maps



Interaction Among Agents

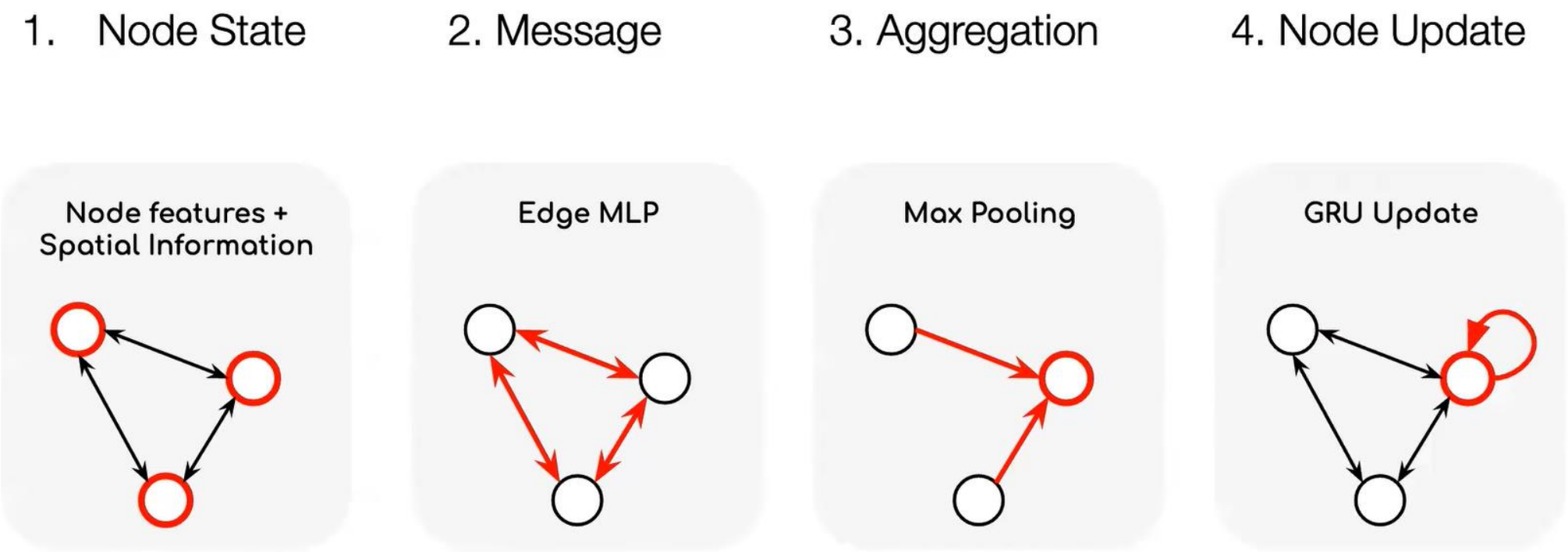
- Interactions is important:
  - Real-world driving is a social activity: agents follow, yield, avoid, and react to each other.
  - To make safe, realistic predictions, the model must understand who interacts with whom and how.
- How to model this:
  - Represent the scene as a graph: Nodes = Agents (with track features + spatial information) Edges = Interaction potential (e.g., spatial proximity, lane connectivity)
  - Use e.g., GNNs, to perform "social reasoning" on this graph.

Encoding Actor Interactions



Interaction: Reasoning in a Graph

- Reasoning in a graph with GNNs



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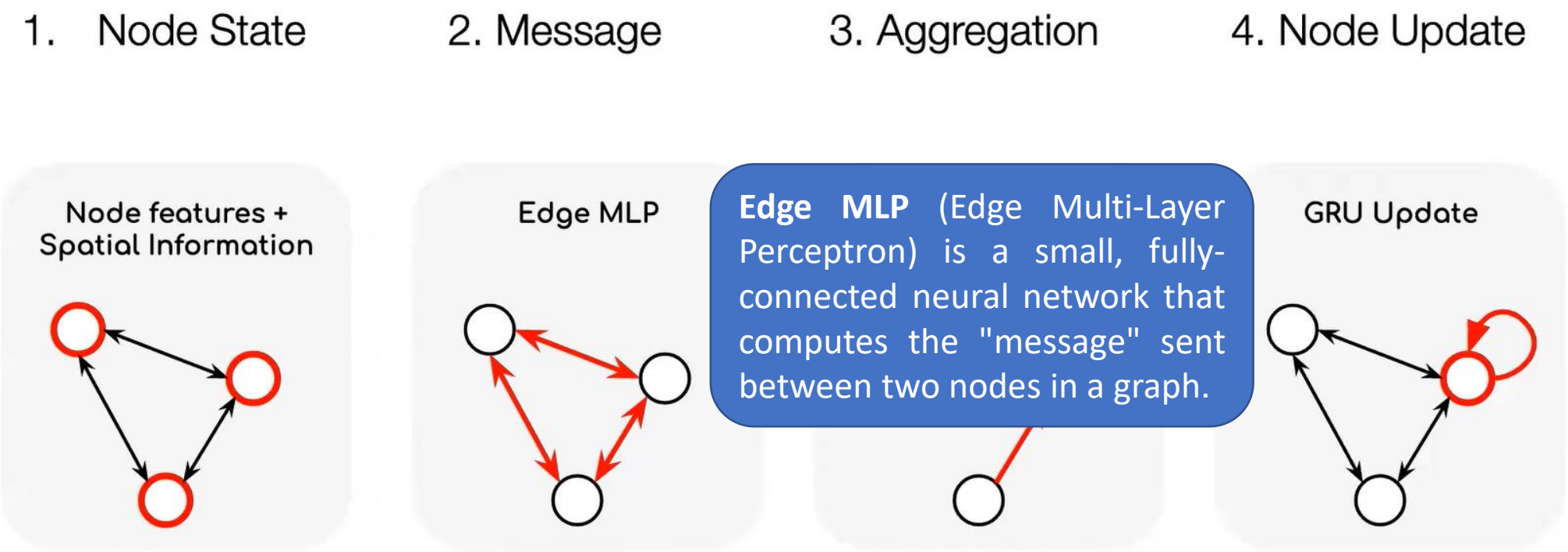
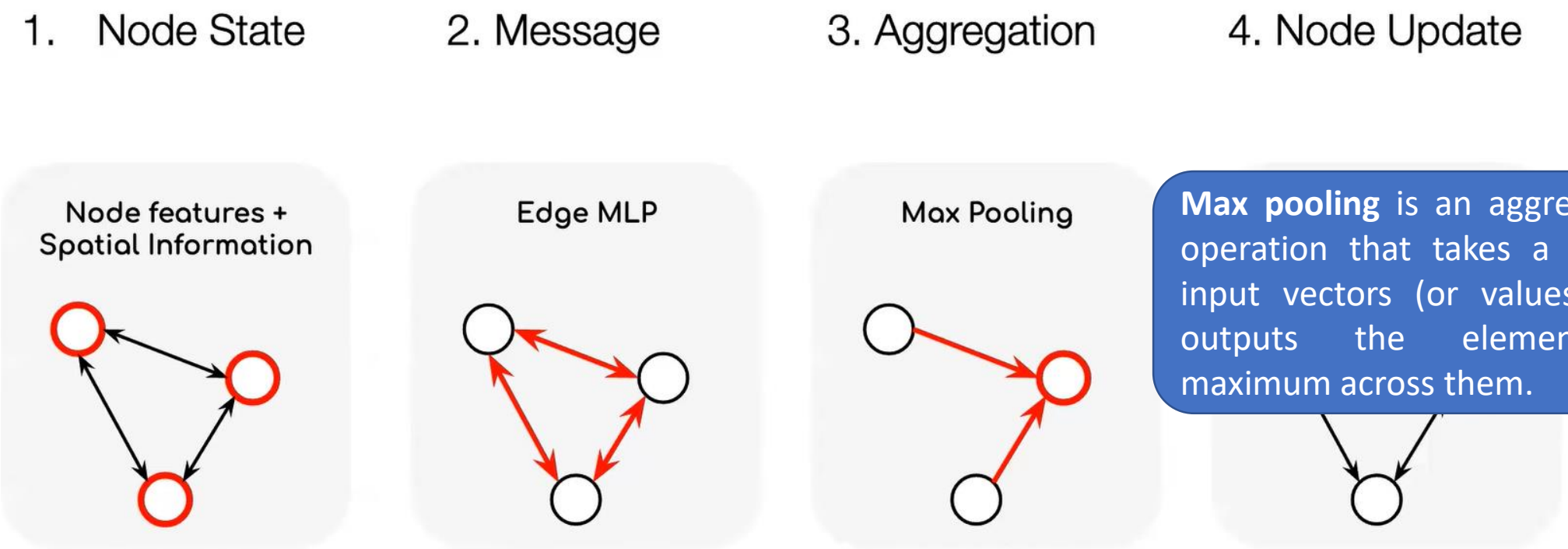


Edge MLP (Edge Multi-Layer Perceptron) is a small, fully-connected neural network that computes the "message" sent between two nodes in a graph.

Interaction: Reasoning in a Graph

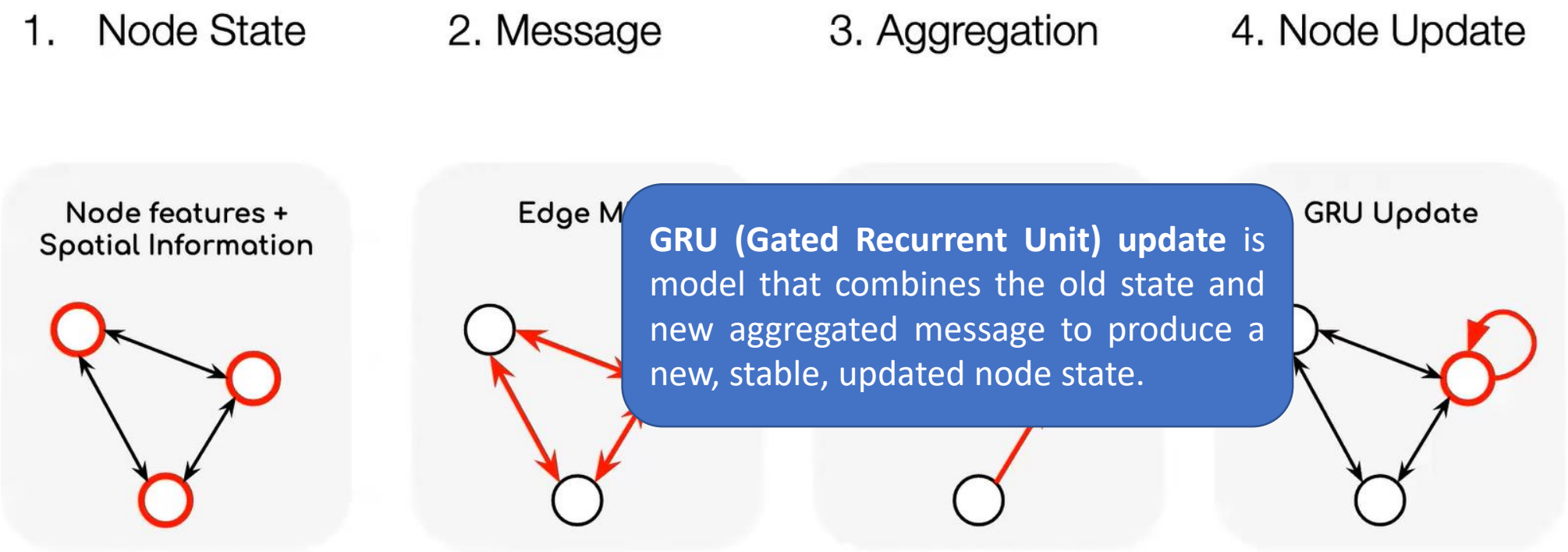
- Reasoning in a graph with GNNs



Max pooling is an aggregation operation that takes a set of input vectors (or values) and outputs the element-wise maximum across them.

Interaction: Reasoning in a Graph

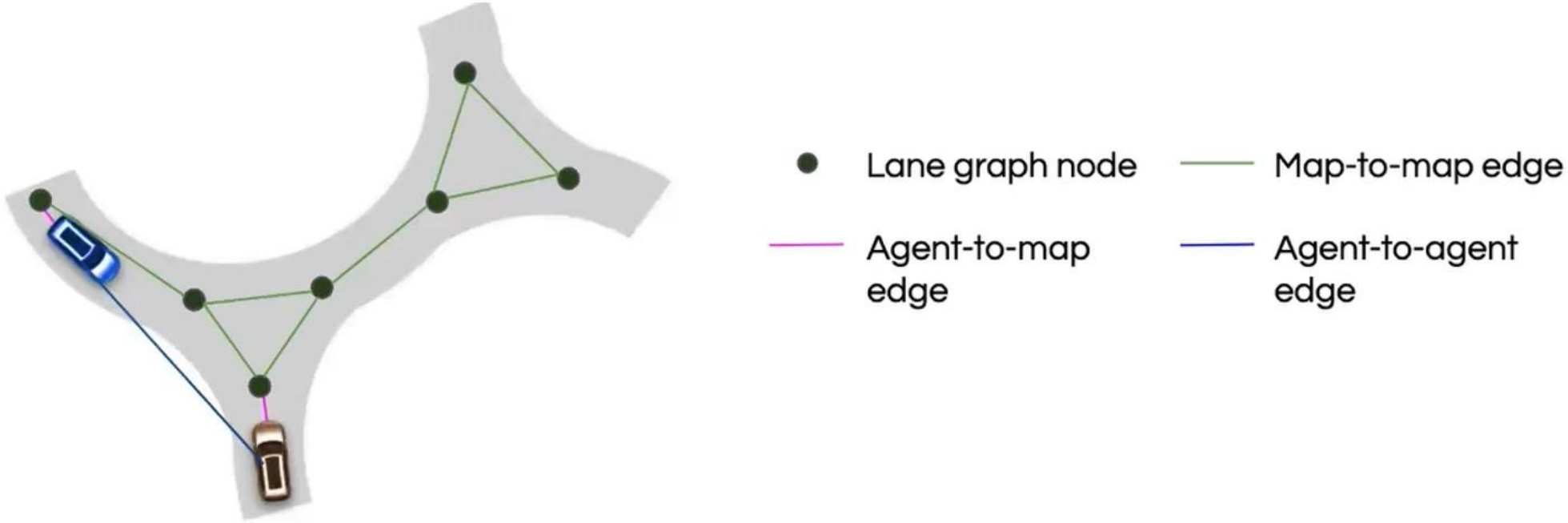
- Reasoning in a graph with GNNs



GRU (Gated Recurrent Unit) update is model that combines the old state and new aggregated message to produce a new, stable, updated node state.

Encoding the Scene as a Graph

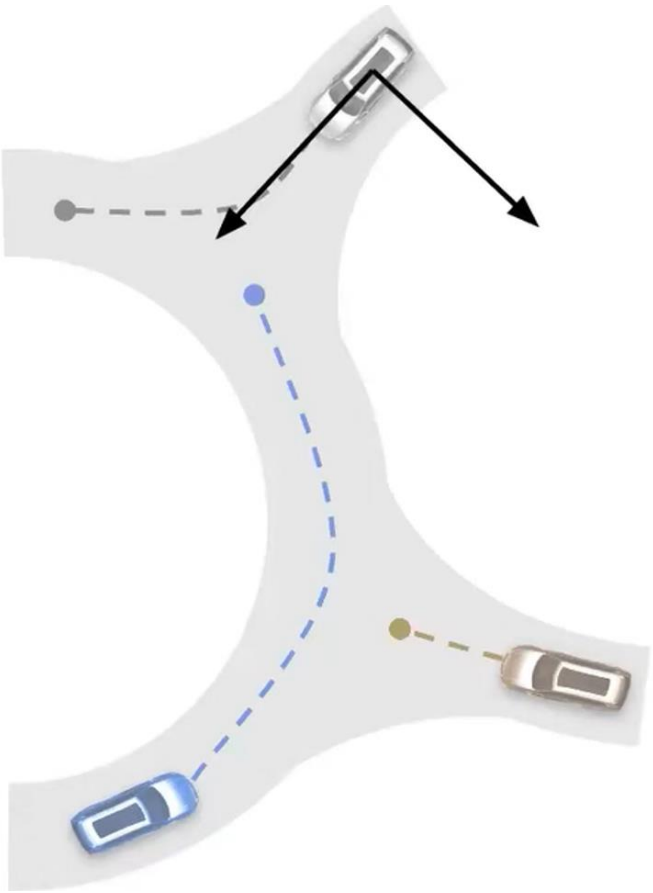
- We can take it beyond agent-to-agent interactions
- Heterogeneous nodes: agent, lane-graph nodes, traffic light states, etc.



Agent-centric vs. Scene-centric Encoding

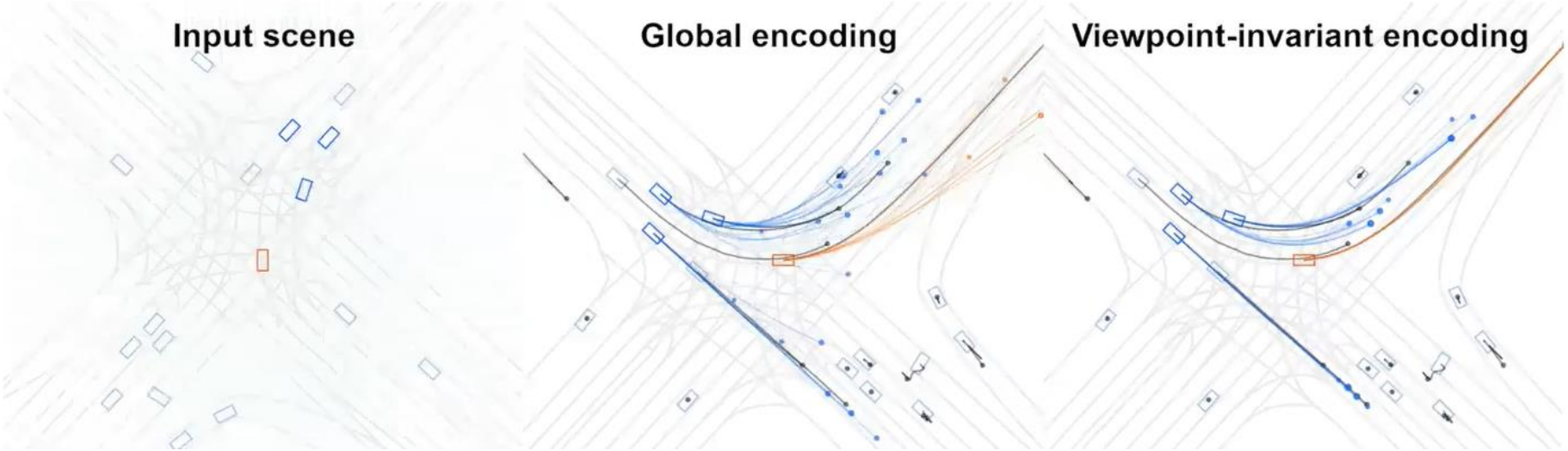
- Agent-centric: Process the scene from the perspective of every agent
  - Slow inference
  - Viewpoint-invariant, sample efficient training

- Scene-centric: Adopt an arbitrary reference frame (e.g., ego vehicle) shared by all agents
  - Fast inference
  - Viewpoint-dependent, sample inefficient



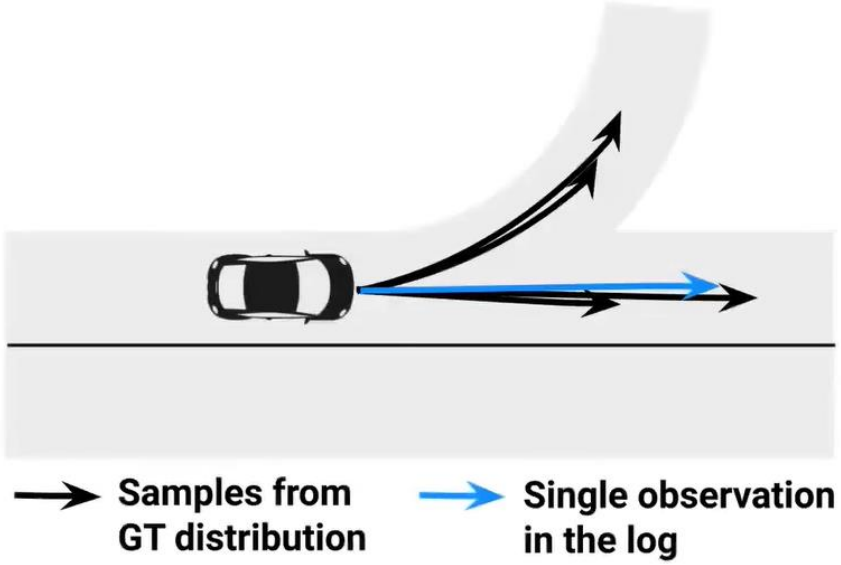
Hybrid Encoding

- Achieve the best of both worlds by Pair-wise Relative Geometric Encoding
  - Node features only contain geometric information in node frame
  - Edge features contain the pose of the sender node in the receiver coordinate frame
- Fast, viewpoint-invariant, and sample efficient



Capturing Multiple Futures

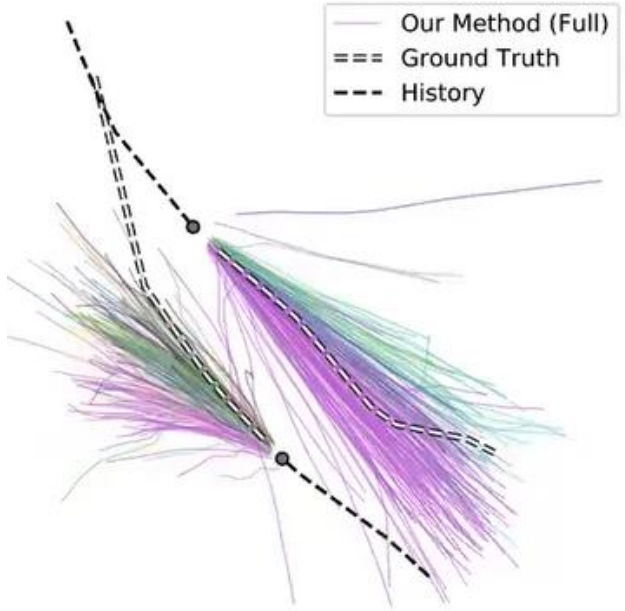
- To plan a safe trajectory, we need to capture the full distribution over other actors future trajectories
- Challenge: Only one sample from the distribution is observed in the dataset
- Solution: Learn aggregate modes from the full dataset



Auto-regressive Decoding

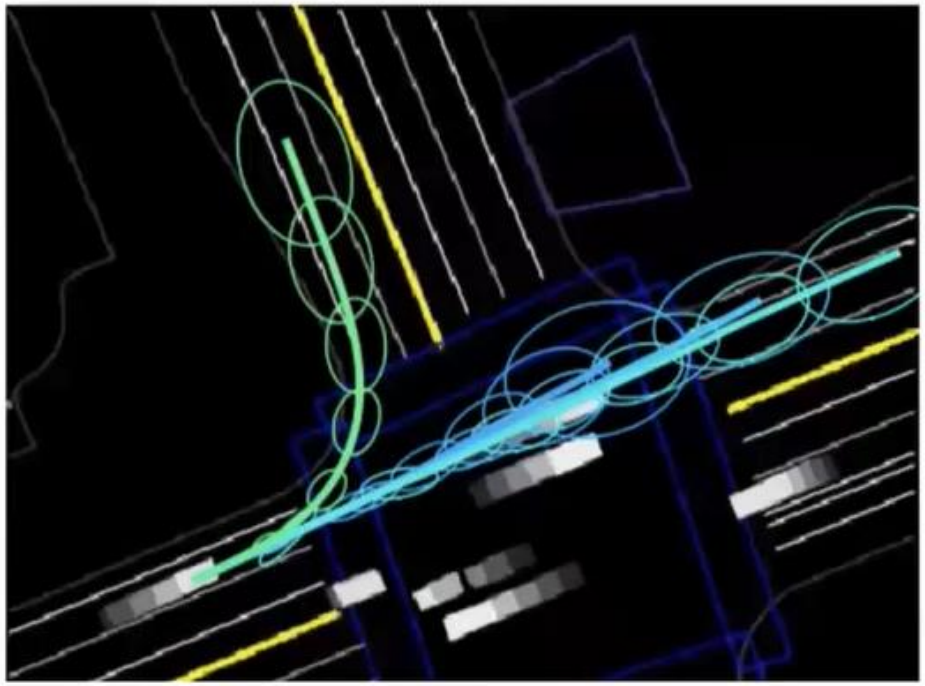
- Model the conditional distributions $p(x_{t+1} \mid x_t, \text{context})$ over time
- Sampled step-by-step, multiple samples at each step, to create diverse future
- Even though only one x_{t+1} is observed, the model is trained to output a distribution with variance (uncertainty) instead of a single point

- Limitations:
  - Sequential sampling is slow
  - Compounding errors (error propagation)



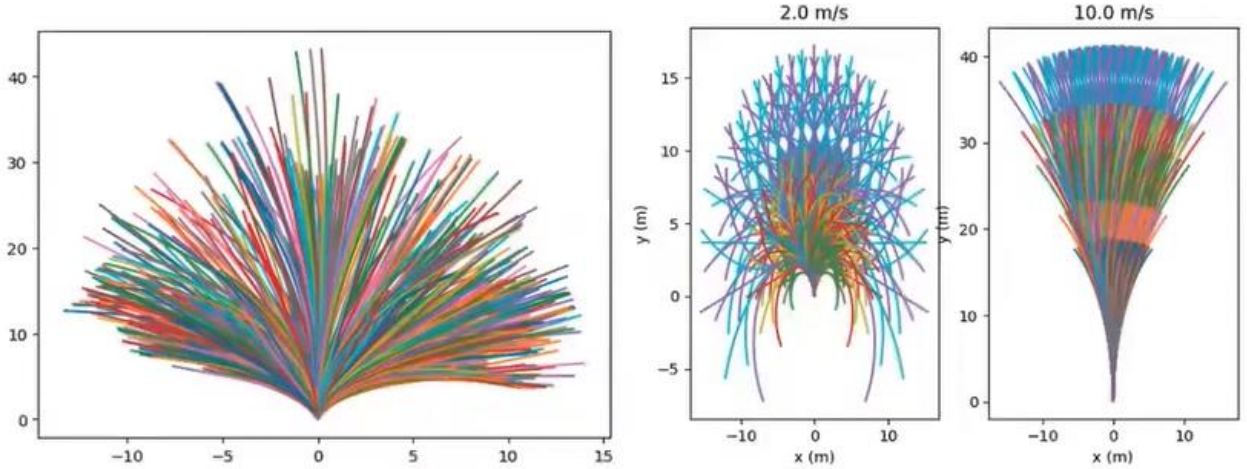
Mixtures of Gaussian Trajectories

- Simple representation to represent multi-modality
  - K mode probabilities, K trajectory regressions with T waypoints each
- Challenge: Probabilistic training results in mode collapse and instabilities



Decoding as Selection in Trajectory Sets

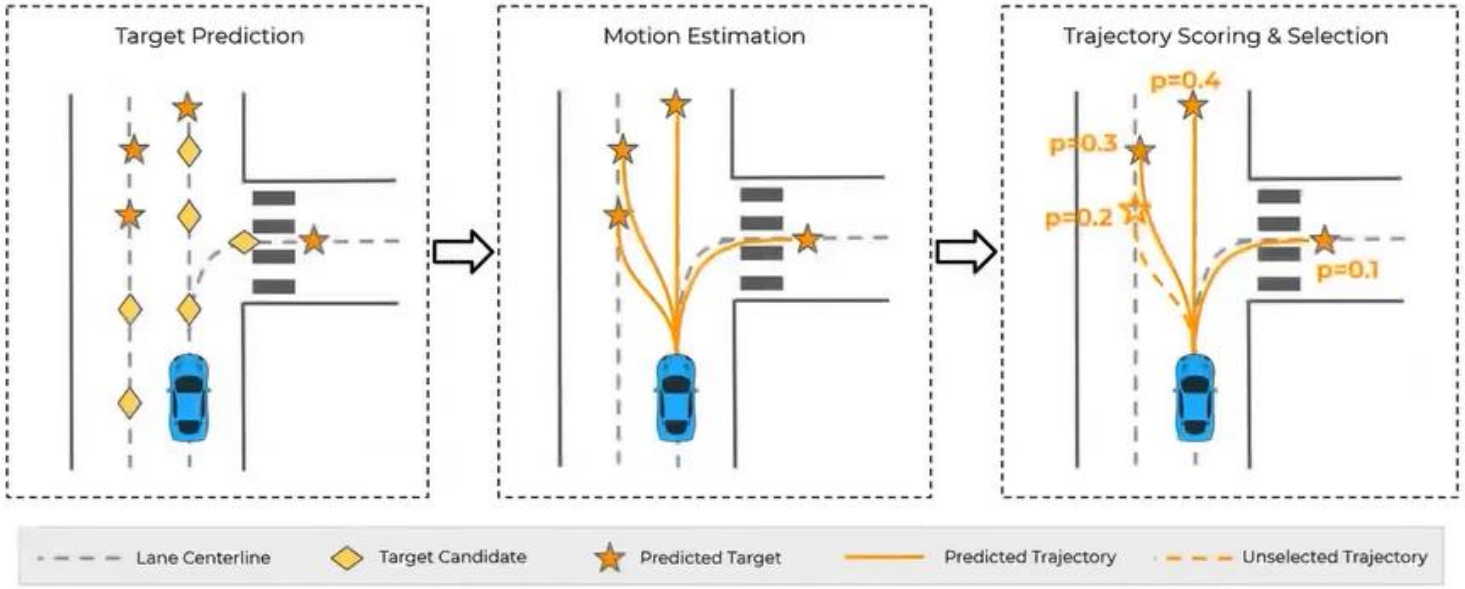
- **Trajectory Set** is a database of real trajectories from the training dataset
  - Can guarantee feasible dynamics
- Multiclass classification loss or energy-based formulation
- Condition on initial kinematic state
- Limited coverage



(a) fixed (b) dynamic

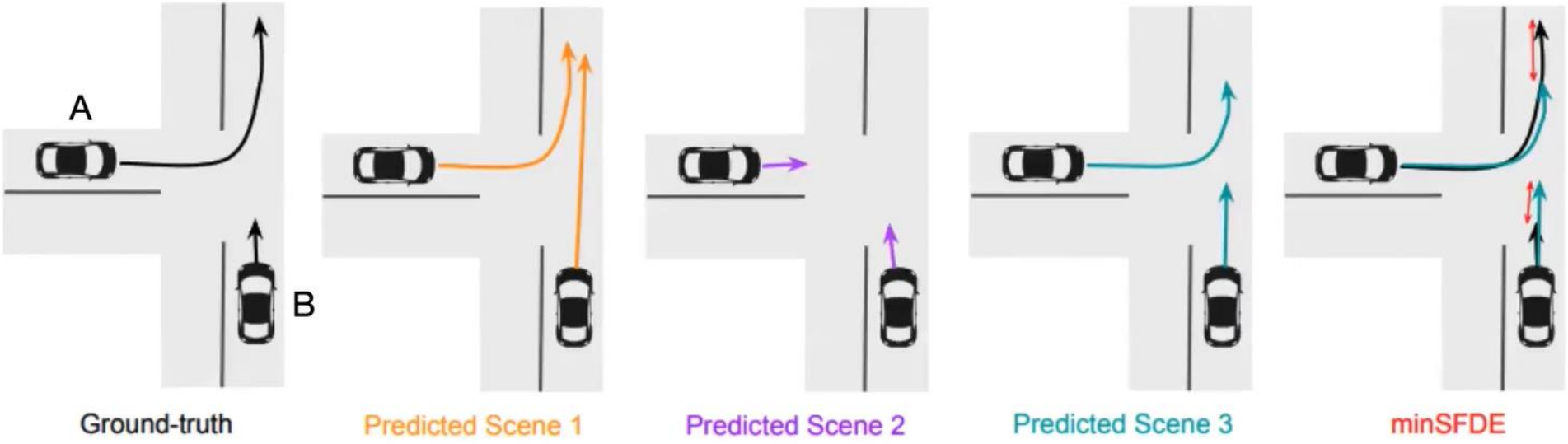
Goal-based Decoding

- Capture future modes with a set of target end points
- Explicit map-relative parameterization aids map understanding
- May not capture non-compliant behaviors



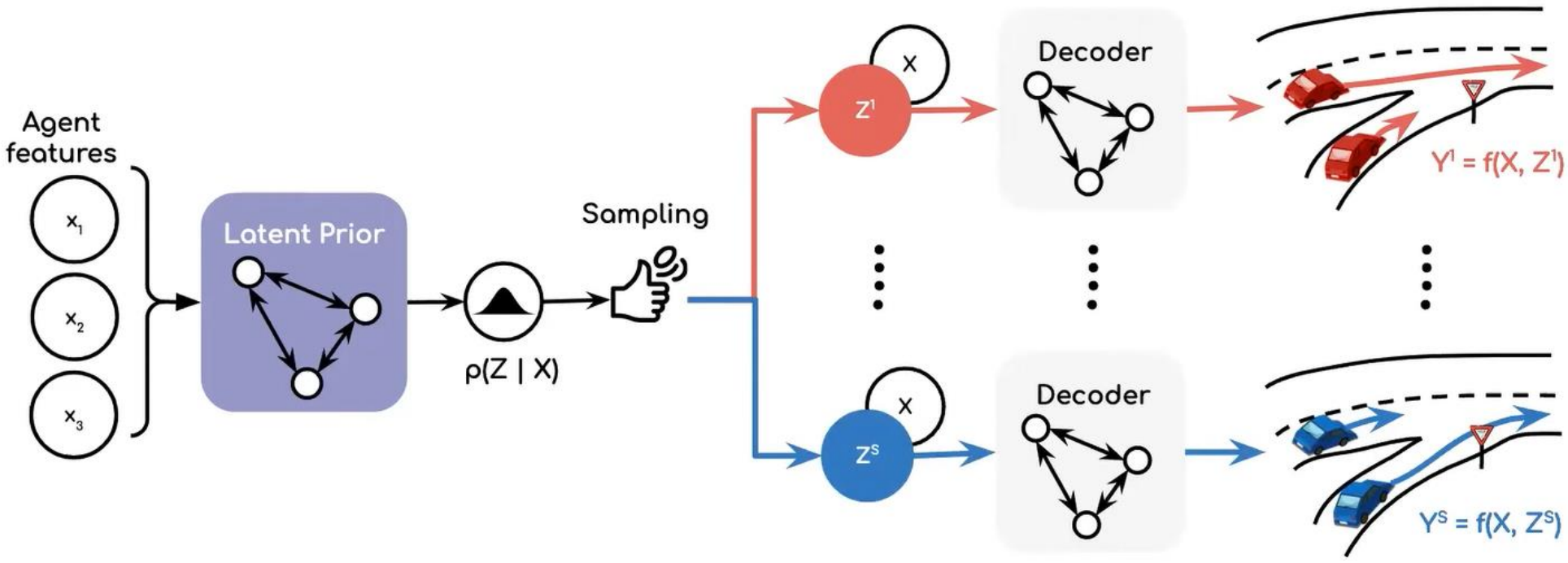
Scene-consistent or Joint Prediction

- Modeling uncertainty per actor does not allow us to understand the distinct futures at the scene level
- Instead we should understand the scene as a whole to plan multiple reactions



Generative Joint Prediction

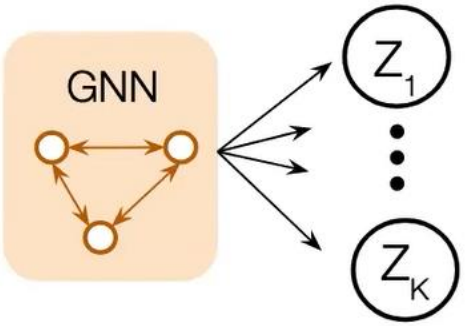
- Scene-level interaction reasoning
- Deterministic decoder with efficient parallel sampling



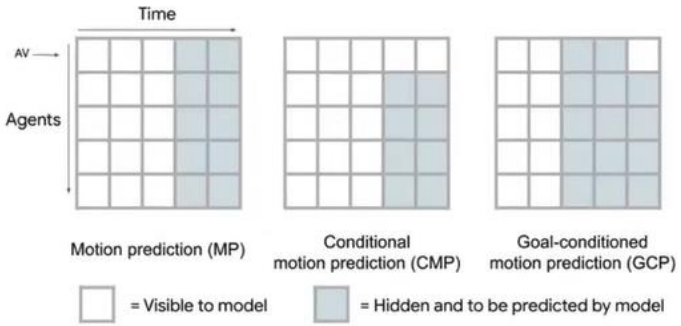
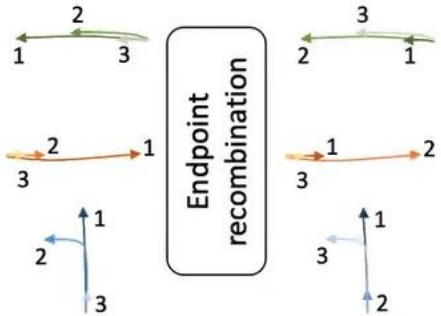
Deterministic Joint Prediction

- Generative models can be hard to deploy in an onboard system
- Models that predict a fixed number of joint modes have been proposed too

Learning a diverse sampler



Masked Sequence Modeling-style



Recombination of marginal modes